

Name: solution (please print)

Signature: \_\_\_\_\_

ECE 2202 – Quiz 2

September 25, 2025

1. Show all work necessary to complete the problem. Use additional sheets of paper as needed.
2. Show all units in solutions and figures. Units in the quiz will be included between square brackets.
3. You will have 40 minutes to work on the quiz.
4. Write your ID number on every page that you show your work.

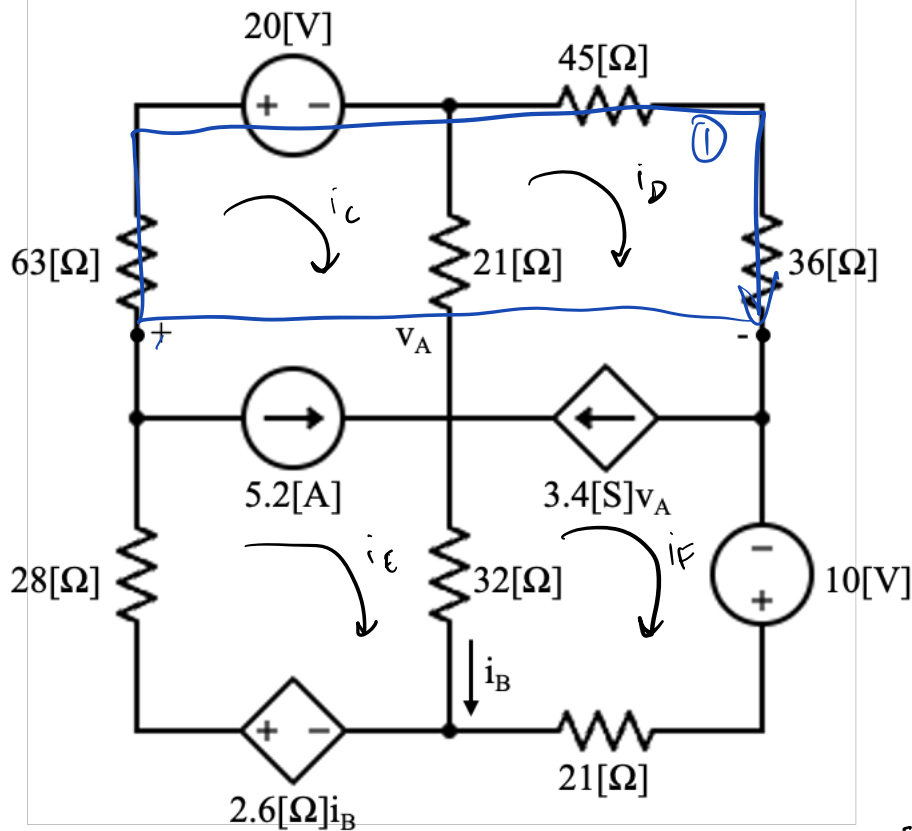
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## Problem 1 (10 pts)

Use the mesh-current method to write a complete set of independent equations that could be used to solve the circuit below. Define all variables appropriately. Do not attempt to simplify the circuit. Do not attempt to simplify or solve your equations.

Labeling (+1 pt)

4 meshes, 2 unknowns  
6 eqns.



$$\begin{aligned}
 & i_C + i_E \quad i_C \cdot 63[\Omega] + 20[V] + (i_C - i_D) \cdot 21[\Omega] + (i_E - i_F) \cdot 32[\Omega] - 2.6[\Omega]i_B + i_E \cdot 28[\Omega] = 0 \quad (+2) \\
 & i_C + i_E \quad i_E - i_C = 5.2[A] \quad (+1) \\
 & i_D + i_F \quad i_D \cdot (45[\Omega] + 36[\Omega]) - 10[V] + i_F \cdot 21[\Omega] + (i_F - i_E) \cdot 32[\Omega] + (i_D - i_C) \cdot 21[\Omega] = 0 \quad (+2) \\
 & i_D + i_F \quad i_D - i_F = 3.4[S]v_A \quad (+1)
 \end{aligned}$$

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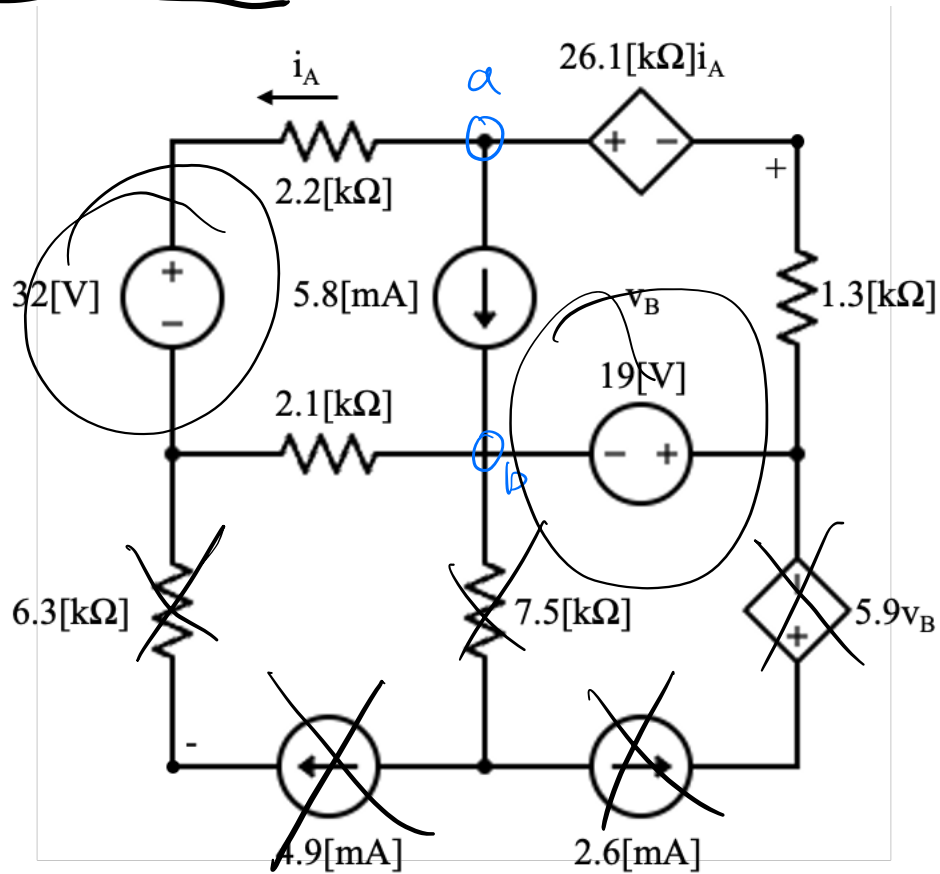
(VA)

$$\text{KVL } \textcircled{1}: -v_A + i_C \cdot 63[\Omega] + 20[V] + i_D \cdot (45[\Omega] + 36[\Omega]) = 0 \quad \textcircled{+2}$$

$$\textcircled{i_B} \quad i_E - i_F = i_B \quad \textcircled{+1}$$

## Problem 2 (10 pts)

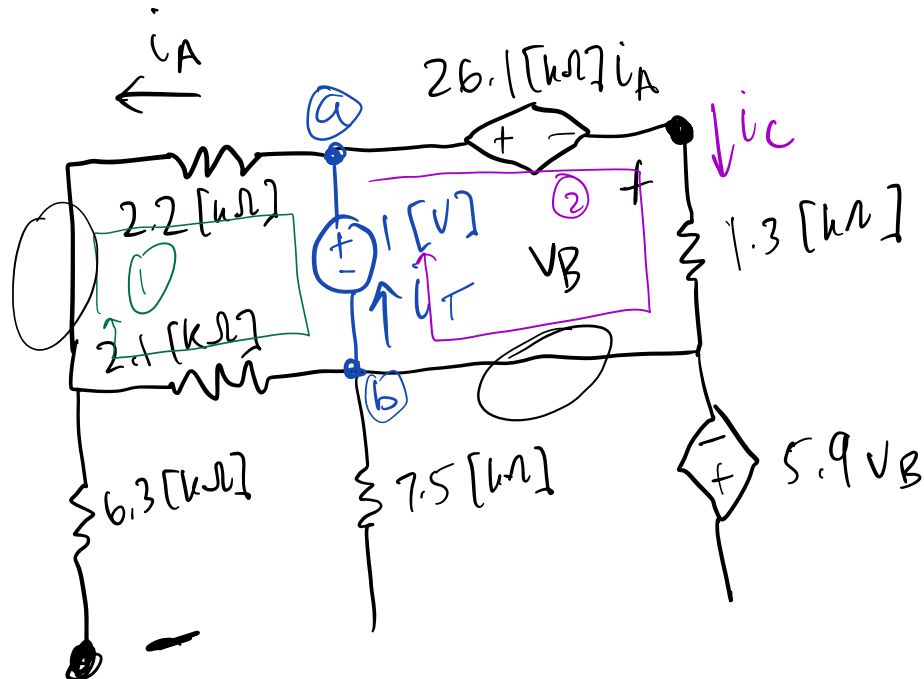
Use circuit given below to solve. Find the Norton equivalent resistance as seen by the 5.8[mA] current source.



see next page

redraw circuit (+3 pts)  
units (+1 pt)

ID number: \_\_\_\_\_



$$R_{eq} = \frac{v_T}{i_T} = \frac{1[V]}{i_T}$$

$$\text{KVL } ① \quad -i_A \cdot 2.2[k\Omega] + 1[V] - i_A \cdot 2.1[k\Omega] = 0 \quad (+2)$$

$$i_A = 0.233 [mA]$$

$$\text{KVL } ② \quad -1[V] + 26.1[k\Omega]i_A + i_C \cdot 1.3[k\Omega] = 0 \quad (+2)$$

$$i_C = -3.91 [mA]$$

$$\text{KCL } (a) \quad i_A - i_T + i_C = 0 \quad (+1)$$

$$i_T = -3.68 [mA]$$

$$R_{eq} = \frac{1[V]}{-3.68 [mA]} = \boxed{-272 [\Omega]} \quad (+1)$$